

APARAJITA GUPTA

San Ramon, California, 94582

☎ 925-290-5330 ✉ gguptaparajita@gmail.com [in linkedin.com/in/guptaaparajita](https://www.linkedin.com/in/guptaaparajita) [globe aparajitawebsite.vercel.app](https://aparajitawebsite.vercel.app)

Education

Purdue University

Graduating May 2028

Bachelor of Science in Computer Engineering, Concentration in AI/ML

West Lafayette, IN

- Activities/Societies: IEEE Motorsports, Data Mine, Embedded Systems @ Purdue, Society of Women Engineers
- Relevant Coursework: Electrical Engineering Fundamentals, Digital System Design, Microprocessor Systems and Interfacing, Signals and Systems, Data Structures, Python for Data Science, Advanced C Programming, Linear Algebra, Multivariate Calculus, Ordinary Differential Equations, Discrete Math

Experience

Inari - Digital Phenotyping with Soybean Imagery

Aug 2025 – May 2026

Data Mine Researcher (ML + Data Science)

Purdue University

- Linked plant phenotypes to genotypes, achieving accuracy of 96.4% and an F1 score of 0.965 using ResNet-18 CNN
- Extracted key physical plant traits by developing phenotyping algorithms to process and analyze 8K soybean images
- Optimized dataset quality by conducting EDA, OTSU/SAM segmentation, and feature extraction (DGCI, MPRI, HSV)
- Identified top predictor by evaluating 5+ ML models in Scikit-learn & TensorFlow (ResNet-18 CNN, KNN, SVM, RF)

IEEE Motorsports

Jan 2025 – Present

Electrical Engineer

Purdue University

- Contributed to electrical design of a 24s8p, 192-cell battery system with 36 backup cells for a student-built EV race car
- Minimized power loss to 5W by integrating nickel strips and BMS with thermal management for safe motor control
- Redesigned automated precharge circuit to operate with 0 chatter or lag, replacing manual switch-based charging
- Earned 1st Place in Battery Efficiency at the EV Grand Prix with a 4,320 Wh, 86.4V battery sustaining a 55-min race

The Data Mine

Aug 2024 – May 2026

Undergraduate Data Science Researcher

Purdue University

- Analyzed 10+ real-world datasets using statistical techniques and visualizations to generate actionable insights
- Performed full-stack data analysis with R, Python, SQL, Bash, Polars, PyTorch, and MongoDB for cleaning/modeling
- Collaborated in a 12-member Agile team, improving deliverable quality & team communication through biweekly sprints
- Delivered technical reports with self-assessments and future goals, delivering findings implemented by industry partners

Projects

3-Band Analog Audio Equalizer | LM356, LM386, RC Filters, Oscilloscope

Analog & Signal Processing

- Designed & built a 5-stage analog equalizer with bass, mid, and treble controls via passive RC filters and op-amps
- Verified the 3 filter cutoff frequencies all well under $\pm 10\%$ tolerance via oscilloscope frequency response analysis
- Achieved < 1.26 mVRMS output ripple across full 100 Hz–10 kHz spectrum, confirming flat and consistent equalization
- Drove an 8Ω speaker at 578–601 mW output via LM386 power amplifier across all tested frequencies

BrightGrade | Python, Playwright, Google Gemini API, Todoist REST API, AppleScript

Backend Automation

- Automated deadline tracking through a dual-source pipeline for 219 tasks across 7 courses, resolving 88% of due dates
- Integrated Gemini API to auto-generate task titles and parse syllabus PDFs, matching user style across 193 tasks
- Fixed 30+ bugs by validating against live account data, correcting 35+ real tasks & reducing false-positive noise by 50%
- Delivered live results via Todoist & AppleScript-driven Calendar.app events, syncing exams & 40+ tasks

IoT Home Sprinkler System | C, Python, Swift, ESP32, REST API

Embedded Systems

- Designed an ESP32 microcontroller-based IoT sprinkler system controlling 4 water zones, with Wi-Fi scheduling
- Programmed logic in C/Python and established Wi-Fi connectivity for real-time remote system management
- Integrated an iOS Swift application via a REST API to enable remote control, achieving < 1 s response latency per zone
- Designed analog LCD interface to display sensor data directly on the device, providing real-time feedback

Technical Skills

Languages: C/C++, C#, Python, Java, Bash, MATLAB, System Verilog

Libraries/Frameworks: NumPy, Pandas, Matplotlib, MongoDB, GitHub, React.js, Keras, Linux/Unix shell scripting

Engineering Skills: Circuit design, electrical wiring, prototyping, KiCAD, IoT device programming, Excel